Personal Statement — MSc in Artificial Intelligence and Robotics (MAIR), The Chinese University of Hong Kong, Shenzhen

The ballbot I worked on at AIRS had to stay balanced while two arms moved on top of it and shifted its centre of mass continuously. Getting it to do so on hardware showed me that every model, state estimate and control output has to survive real constraints, and that the tools I relied on, linear models and hand-set gains, have limits I could not push past without more mathematics, optimisation and learning. I am applying to the MSc in Artificial Intelligence and Robotics to build those foundations and turn practical experience into systematic algorithm design.

I am a final-year Computer Science student at Shenzhen Technology University, ranked 6 of 156, and I took Principles of Automatic Control and Modern Control Theory in addition to the CS curriculum. At the Shenzhen Institute of Artificial Intelligence and Robotics for Society, I was responsible for the balance control of a three-omni-wheel ballbot carrying two arms. I designed and deployed a cascaded PID controller, then modelled the system as a linearised four-state inverted pendulum and used LQR to understand how state weights trade stability against control effort, which explained the gain tuning and actuator saturation I was seeing on hardware. I estimated the arms' centre-of-mass shift with URDF/Pinocchio and fed it forward as gravity compensation. The chassis maintained dynamic balance while both arms moved along fixed trajectories.

Two laboratory projects took me from balance to locomotion and navigation. On a wheel-legged robot with variable leg length I implemented gain-scheduled LQR and virtual model control through closed-chain leg kinematics and tuned it on the real machine. On a car-like mobile robot I built the chain from EKF/AMCL localisation through Hybrid-A* and TEB planning to MPC tracking and ran it on the vehicle. At Future Power I moved to manipulation: VR teleoperation of a dual-arm robot, solving constrained inverse kinematics in MuJoCo/Mink so the arms follow the operator without velocity spikes near singularities, and a leader–follower controller with gravity compensation, built on URDF/KDL dynamics, for hand-guided teaching. Earlier, I was responsible for the embedded control of a RoboMaster hero robot that reached the 2025 national final, and built a personal OpenHarmony project that closed a loop from edge vision inference through a LangGraph decision chain to an actuator.

What these projects share is that I built each model for a specific robot and tuned each controller against its hardware. I have not yet had a formal course in manipulation, optimisation as a design tool or reinforcement learning, and I want to reason about these systems with optimisation and probability. This is why I am choosing the Robotics concentration. Robot Manipulation and Math Fundamentals for Robotics are exactly the foundations my dual-arm work lacked; the core courses in Artificial Intelligence and Machine Learning will give me a principled basis for the learning-based perception I have so far only deployed; and the electives in Mobile Robotics, Reinforcement Learning and Optimization map directly onto my navigation and control experience.

My goal is to work as a robotics algorithm engineer in the Greater Bay Area, on navigation, control and learning for mobile and dual-arm robots, and to grow into someone who can own the algorithm stack of a robot product. The programme’s combination of mathematical foundations, machine learning and embodied intelligence, and its position inside Shenzhen's robotics industry, where I have interned at a research institute and two companies, make it a strong fit for my goals. My preparation is practical: I have carried control, navigation and manipulation modules through hardware validation, and I know

which theory I still need. I intend to use the two years to build that foundation systematically.